Fraud/Scam Analytics Data Mart

Consolidated Data Dictionary

Version 0.2 - Phase 8.1 Approved Blueprint Baseline

Document field	Value
Project	Fraud/Scam Analytics Data Mart
Current phase	Phase 8 - Gold Layer Build
Completed milestone	Phase 8.1 - Gold Model Blueprints
Primary fact grain	One row per fraud/scam investigation case
Primary fact table	fct_scam_cases
Status	Approved modelling baseline for physical implementation

Purpose of this document
Provide one consistent reference for Raw, Silver and Gold table definitions, key strategies, unknown-member handling, measures, tests and outstanding implementation decisions. Use it during dbt and Snowflake build work, validation, Power BI modelling and Phase 11 documentation.

Contents

1. Modelling standards and approved project decisions
2. Gold-layer tables
   2.1 fct_scam_cases
   2.2 dim_customer
   2.3 dim_account
   2.4 dim_product
   2.5 dim_channel
   2.6 dim_scam_type
   2.7 dim_rule
   2.8 dim_date
3. Raw-layer tables
4. Silver-layer tables
5. Source-to-target summary
6. Decision register and remaining implementation items

Version 0.2 change summary

- Updated all Gold-table statuses to reflect completion of Phase 8.1 blueprints.
- Expanded fct_scam_cases with full unknown-member mapping, date handling, physical measures, reporting metrics and test requirements.
- Added dim_date, including Unknown Date and Not Applicable members.
- Confirmed deterministic hashed surrogate keys for business dimensions and case_key.
- Confirmed YYYYMMDD integer keys for normal date rows, with reserved special date keys.
- Clarified current-state handling for dim_customer and dim_account; SCD Type 2 remains a future enhancement.
- Separated No Rule Triggered from Unknown Rule in dim_rule.
- Clarified that descriptive dates in dimensions may be NULL without breaking fact-to-dimension referential integrity.
- Retained Raw and Silver sections as working physical definitions pending workbook reconciliation.

1. Modelling standards and approved project decisions

Layer	Purpose
Raw	Preserves source-system data with minimal transformation and retains realistic data-quality problems.
Silver	Cleans, standardises, casts, deduplicates and prepares source data. Uses natural keys and quality flags.
Gold	Creates reporting-ready dimensions and facts. Introduces surrogate keys, business measures and controlled special members.

1.1 Approved standards

- The business problem is to provide trusted reporting for fraud/scam cases, financial loss, detection effectiveness and operational performance.
- fct_scam_cases is the main fact table and contains one row per fraud/scam investigation case.
- The model is case-level first. A transaction-level fact is not being built in this version.
- Transaction and alert impact is summarised at case grain using transaction_count and alert_count.
- Natural keys originate from source systems and remain available in Silver and Gold dimensions.
- Surrogate keys are created only in Gold.
- Business dimensions use deterministic hashed surrogate keys generated from standardised natural keys.
- The controlled natural key UNK generates Unknown-member surrogate keys.
- dim_rule also uses NO_RULE for valid cases where no automated rule triggered.
- dim_date uses YYYYMMDD integer keys for real dates, 0 for Unknown Date and -1 for Not Applicable.
- Fact-table foreign keys should not be NULL; invalid, missing or unmatched references map to controlled special members.
- A missing case_id is not mapped to an Unknown Case. The row must be rejected or quarantined because the grain cannot be protected.
- dim_customer and dim_account are current-state dimensions. SCD Type 2 is a future enhancement.
- Measures must match the fact grain. Reporting measures calculated across rows are not stored repeatedly on each fact row.
- Gold should be business-ready; technical quality flags normally remain in Silver or dedicated quality models unless reporting requires them.

1.2 Null and special-member rules

Scenario	Approved treatment
Missing fact-table dimension reference	Map to the corresponding Unknown dimension surrogate key.
Valid case with no automated rule	Map rule_key to the NO_RULE member, not the Unknown Rule member.
Open case with no closure date	Map case_closed_date_key to Not Applicable (-1).
Closed case with missing closure date	Map case_closed_date_key to Unknown Date (0) and raise a quality issue.
Descriptive date attribute in a dimension is genuinely unknown	NULL is acceptable because the fact relationship uses the dimension surrogate key, not the descriptive date attribute.
Boolean value is genuinely unknown	Use NULL rather than FALSE, because FALSE means a confirmed negative.
Measure equals zero	Use 0 only when the business meaning is confirmed none.
Measure is unavailable or not applicable	Use NULL where the distinction from zero matters.

2. Gold-layer tables

Gold-layer objective

Provide a simple reporting-ready star schema in which fct_scam_cases remains one row per case and every dimension lookup preserves that grain.

2.1 fct_scam_cases

Status: Approved Phase 8.1 blueprint; ready for physical implementation after environment and Silver setup.

Business purpose: Captures case-level fraud/scam investigation records and serves as the central analytical fact for case volumes, financial exposure, prevention, recovery, reimbursement, detection outcomes and operational performance across customers, accounts, products, channels, scam types, rules and time.

Grain: One row per unique fraud/scam investigation case_id.

Primary Silver source: stg_scam_cases

Natural key: case_id

Hashed surrogate key: case_key, deterministically generated from standardised case_id.

Keys and case attributes

Column	Role	Definition
case_key	Hashed surrogate key	Deterministic Gold key generated from case_id.
case_id	Natural key	Source investigation case identifier. Defines and protects the fact grain.
customer_key	Foreign key	References dim_customer.customer_key.
account_key	Foreign key	References dim_account.account_key.
product_key	Foreign key	References dim_product.product_key.
channel_key	Foreign key	References dim_channel.channel_key.
scam_type_key	Foreign key	References dim_scam_type.scam_type_key.
rule_key	Foreign key	References the primary or originating rule in dim_rule.
case_created_date_key	Foreign key	References dim_date for case creation.
case_closed_date_key	Foreign key	References dim_date for closure, Unknown Date or Not Applicable.
loss_occurred_date_key	Foreign key	References dim_date for loss occurrence, Unknown Date or Not Applicable.
case_status	Degenerate attribute	Standardised current investigation status.
case_outcome	Degenerate attribute	Standardised current or final outcome.
case_priority	Degenerate attribute	Operational investigation priority.
is_confirmed_scam	Indicator	1 when the case meets the confirmed-scam definition; otherwise 0.
is_false_positive	Indicator	1 when a closed case meets the false-positive definition; otherwise 0.
is_closed_case	Indicator	1 when the case is closed; otherwise 0.

Physical measures stored at case grain

Column	Definition	Default aggregation
alert_count	Number of alerts associated with the case.	Sum
transaction_count	Number of transactions associated with the case.	Sum
gross_loss_amount	Total financial loss before recovery or reimbursement.	Sum
prevented_amount	Amount stopped before it became an	Sum

Column	Definition	Default aggregation
	actual loss.	
recovered_amount	Amount recovered after the loss occurred.	Sum
reimbursed_amount	Amount reimbursed to the customer.	Sum
net_loss_amount	gross_loss_amount - recovered_amount - reimbursed_amount. Prevented amount is not subtracted.	Sum
days_to_resolve	Days between creation and closure. Populated only for closed cases with valid dates.	Average for resolution analysis

Reporting measures calculated from the fact

Metric	Definition
Total Cases	Count of unique case_id.
Confirmed Scam Cases	Count of cases where is_confirmed_scam = 1.
Closed Cases	Count of cases where is_closed_case = 1.
False Positive Cases	Count of closed cases where is_false_positive = 1.
False Positive Rate	False Positive Cases divided by Closed Cases.
Gross Loss Amount	Sum of gross_loss_amount.
Prevented Amount	Sum of prevented_amount.
Recovered Amount	Sum of recovered_amount.
Reimbursed Amount	Sum of reimbursed_amount.
Net Loss Amount	Sum of net_loss_amount.
Average Days to Resolve	Average days_to_resolve for closed cases only.

Unknown and special-member handling

- The fact table does not contain a seeded Unknown Fact row.
- Every accepted row requires a valid, unique case_id. Missing case_id values are rejected or quarantined.
- Missing, invalid, unmatched or late-arriving dimension natural keys map to the corresponding dimension Unknown surrogate key.
- A valid case with no automated rule maps to dim_rule.NO_RULE; an invalid or missing rule maps to dim_rule.UNK.

Source condition	Fact-key treatment
Missing or unmatched customer_id	Map to dim_customer key generated from UNK.
Missing or unmatched account_id	Map to dim_account key generated from UNK.
Missing or unmatched product identifier	Map to dim_product key generated from UNK.
Missing or unmatched channel_code	Map to dim_channel key generated from UNK.
Missing or unmatched scam_type_code	Map to dim_scam_type key generated from UNK.
Invalid or unmatched rule_id	Map to Unknown Rule key generated from UNK.
Valid case with no automated rule	Map to No Rule Triggered key generated from NO_RULE.
Missing/invalid case_created_date	Map to Unknown Date (0); high-severity quality issue.
Open case with no closure date	Map to Not Applicable (-1).
Closed case with missing/invalid closure date	Map to Unknown Date (0); quality issue.
No financial loss occurred	Map loss_occurred_date_key to Not Applicable (-1).
Loss occurred but date unavailable	Map loss_occurred_date_key to Unknown Date (0).

Required tests

Test area	Required checks
Grain and identifiers	case_key unique/not null; case_id unique/not null.
Foreign keys	All dimension keys not null and relationships-valid. Foreign keys are repeatable and must not be tested as unique.

Test area	Required checks
Indicators	is_confirmed_scam, is_false_positive and is_closed_case restricted to 0/1; false positives must be closed; confirmed and false-positive flags should not both be 1 unless explicitly allowed.
Counts	alert_count and transaction_count non-negative when populated.
Financials	Financial amounts non-negative when populated; net_loss_amount formula validated; any recovery/reimbursement cap rule requires business confirmation.
Lifecycle dates	Closed date not before created date; open cases use Not Applicable and days_to_resolve is NULL; closed cases have non-negative days_to_resolve when valid dates exist.
Accepted values	Approved domains for case_status, case_outcome and case_priority.
Reconciliation	Gold accepted case count and financial totals reconcile to stg_scam_cases; joins do not multiply cases.
Unknown monitoring	Track rates of cases mapped to each Unknown member; thresholds handled in Phase 9.

2.2 dim_customer

Status: Approved Phase 8.1 blueprint.

Business purpose: Provides current customer demographic, regional, segmentation and risk context for analysing scam targeting, financial vulnerability, repeat victimisation and case outcomes across customer groups.

Grain: One row per current standardised customer_id.

Silver source: stg_customers

Natural key: customer_id

Hashed surrogate key: customer_key, generated from customer_id.

History strategy: Current-state dimension. Historical attribute tracking through SCD Type 2 is deferred.

Column	Definition
customer_key	Deterministic hashed Gold surrogate key.
customer_id	Source-system customer identifier.
customer_type	Customer classification, such as Individual or Business.
customer_segment	Business-defined customer segment.
age_group	Reporting-friendly age band.
risk_rating	Current customer risk classification.
state	Standardised customer state or region.
customer_since_date	Date the customer relationship began; may be NULL when genuinely unknown.
customer_status	Current customer status.

Unknown member: customer_id = 'UNK'; customer_key = deterministic hash of 'UNK'; descriptive text attributes = 'Unknown'; customer_since_date = NULL. Null, blank, invalid, unmatched or late-arriving customer IDs map to this member.

Test area	Required checks
Keys	customer_key unique/not null; customer_id unique/not null.
Attributes	Accepted values for controlled domains after confirmation; customer_type not null where required.
Dates	customer_since_date is a valid DATE when populated and <= current_date.

Test area	Required checks
Special member	Exactly one customer_id = UNK with approved defaults.
Fact relationship	fct_scam_cases.customer_key not null and references dim_customer.customer_key.
Grain protection	Customer lookup must not duplicate case_id.

2.3 dim_account

Status: Approved Phase 8.1 blueprint.

Business purpose: Provides current account status, lifecycle-date, account-type and ownership context for analysing account vulnerability, new/dormant account exploitation, financial exposure and operational actions.

Grain: One row per current standardised account_id.

Silver source: stg_accounts

Natural key: account_id

Hashed surrogate key: account_key, generated from account_id.

History strategy: Current-state dimension. Historical account-status tracking through SCD Type 2 is deferred.

Column	Definition
account_key	Deterministic hashed Gold surrogate key.
account_id	Source account identifier.
customer_id	Source customer reference retained for lineage/reconciliation; not used to snowflake Power BI relationships.
product_id or product_code	Source product reference retained pending final physical naming decision.
account_type	Standardised account type.
account_status	Current account status.
account_open_date	Date the account opened; may be NULL when unknown.
account_closed_date	Date the account closed; NULL for open accounts or unknown values.
is_joint_account	Boolean joint-account indicator; NULL when genuinely unknown.
last_updated_at	Optional technical audit metadata.

Unknown member: account_id = 'UNK'; account_key = deterministic hash of 'UNK'; text attributes = 'Unknown'; account_open_date/account_closed_date/is_joint_account/last_updated_at = NULL.

Test area	Required checks
Keys	account_key unique/not null; account_id unique/not null.
Attributes	account_type not null where required; accepted values for account_type and account_status after confirmation; Boolean validation for is_joint_account.
Dates	When both populated, account_closed_date >= account_open_date.
Special member	Exactly one account_id = UNK.
Fact relationship	fct_scam_cases.account_key not null and references dim_account.account_key.
Grain protection	Account lookup must not duplicate case_id.

Known limitation
A joint account may have multiple customers. Retaining one customer_id can only represent the primary/source-provided customer. A future account-customer bridge would be required for complete many-to-many ownership analysis.

2.4 dim_product

Status: Approved blueprint subject to final physical natural-key naming reconciliation.

Business purpose: Provides standardised product context for comparing case volumes, confirmed scam rates, financial losses, prevented amounts and operational outcomes across banking products and product categories.

Grain: One row per standardised banking product identifier.

Silver source: stg_products

Natural key: product_id or product_code - one canonical physical name must be confirmed before implementation.

Hashed surrogate key: product_key, generated from the confirmed standardised product natural key.

Column	Definition
product_key	Deterministic hashed Gold surrogate key.
product_id or product_code	Stable source-system product identifier; final physical name pending workbook reconciliation.
product_name	Business-friendly product name.
product_category	Higher-level product grouping.
product_status	Current product lifecycle status.

Unknown member: Confirmed natural key = 'UNK'; product_key = deterministic hash of 'UNK'; product_name = 'Unknown Product'; product_category/product_status = 'Unknown'.

Test area	Required checks
Keys	product_key unique/not null; product natural key unique/not null.
Attributes	product_name not null; accepted values for product_status and controlled category domains after confirmation.
Special member	Exactly one product natural key = UNK.
Fact relationship	fct_scam_cases.product_key not null and references dim_product.product_key.
Grain protection	Product lookup must not duplicate case_id.

2.5 dim_channel

Status: Approved Phase 8.1 blueprint.

Business purpose: Provides standardised channel context for analysing case volumes, confirmed scam rates, false-positive rates, financial losses and prevention performance across digital, physical and assisted channels.

Grain: One row per standardised channel_code.

Silver source: stg_channels

Natural key: channel_code

Hashed surrogate key: channel_key, generated from channel_code.

Column	Definition
channel_key	Deterministic hashed Gold surrogate key.
channel_code	Standardised source-system channel identifier.
channel_name	Business-friendly channel name.
channel_type	Broad operational classification.
channel_group	Higher-level reporting group.

Unknown member: channel_code = 'UNK'; channel_key = deterministic hash of 'UNK'; channel_name = 'Unknown Channel'; channel_type/channel_group = 'Unknown'.

Test area	Required checks
Keys	channel_key unique/not null; channel_code unique/not null.
Attributes	Accepted values for channel_type/channel_group after business

Test area	Required checks
	domain confirmation.
Special member	Exactly one channel_code = UNK with approved defaults.
Fact relationship	fct_scam_cases.channel_key not null and references dim_channel.channel_key.
Grain protection	Channel lookup must not duplicate case_id.

2.6 dim_scam_type

Status: Approved Phase 8.1 blueprint.

Business purpose: Provides standardised scam classifications for analysing case volumes, confirmed scam rates, financial losses, prevented amounts, resolution performance and emerging scam trends.

Grain: One row per standardised scam_type_code.

Silver source: stg_scam_types

Natural key: scam_type_code

Hashed surrogate key: scam_type_key, generated from scam_type_code.

Column	Definition
scam_type_key	Deterministic hashed Gold surrogate key.
scam_type_code	Standardised source-system scam type identifier.
scam_type_name	Business-friendly scam type name.
scam_category	Broader scam classification.
scam_description	Business description of the scam type.
active_flag	Indicates whether the scam type is currently available for classification. Inactive values remain for historical reporting.

Excluded Gold quality flags: is_missing_scam_category and is_missing_scam_description remain in Silver or a quality model unless a reporting requirement is approved.

Unknown member: scam_type_code = 'UNK'; scam_type_key = deterministic hash of 'UNK'; scam_type_name = 'Unknown Scam Type'; scam_category = 'Unknown'; scam_description = 'Unknown Scam Type Description'; active_flag = agreed inactive/default value.

Test area	Required checks
Keys	scam_type_key unique/not null; scam_type_code unique/not null.
Attributes	Accepted values for scam_category and active_flag after domain confirmation.
Special member	Exactly one scam_type_code = UNK; default attributes match approved values.
Fact relationship	fct_scam_cases.scam_type_key not null and references dim_scam_type.scam_type_key.
Grain protection	Scam-type lookup must not duplicate case_id.

2.7 dim_rule

Status: Approved Phase 8.1 blueprint.

Business purpose: Provides standardised rule context for comparing case volumes, confirmed scam rates, false-positive rates, losses, prevented amounts and rule effectiveness.

Grain: One row per current standardised rule_id.

Silver source: stg_rules

Natural key: rule_id

Hashed surrogate key: rule_key, generated from rule_id.

Fact relationship meaning: fct_scam_cases.rule_key represents the primary or originating detection rule. It does not represent every contributing alert or rule.

Column	Definition
rule_key	Deterministic hashed Gold surrogate key.
rule_id	Standardised source-system rule identifier.
rule_name	Business-friendly rule name.
rule_category	Higher-level detection-rule grouping.
rule_description	Description of the rule purpose or logic.
is_active	Indicates whether the current rule definition is active.
rule_version	Current version retained as a descriptive attribute.
rule_start_date	Current rule effective start date.
rule_end_date	Current rule end date when applicable.
last_updated_date	Optional technical audit metadata.

Special member	Controlled values and meaning
No Rule Triggered	rule_id = 'NO_RULE'; rule_key = hash('NO_RULE'); rule_name = 'No Rule Triggered'; rule_category = 'Manual'. Valid manual/customer-reported case.
Unknown Rule	rule_id = 'UNK'; rule_key = hash('UNK'); rule_name = 'Unknown Rule'; rule_category = 'Unknown'. Missing, invalid, unmatched or late-arriving rule reference.

Test area	Required checks
Keys	rule_key unique/not null; rule_id unique/not null.
Attributes	Accepted values for rule_category and is_active after domain confirmation.
Dates	When populated, rule_end_date >= rule_start_date.
Special members	Exactly one NO_RULE record and one UNK record.
Fact relationship	fct_scam_cases.rule_key not null and references dim_rule.rule_key.
Grain protection	Rule lookup must not duplicate case_id.

Future enhancement
If cases identify the applicable rule version and historical rule-version analysis is required, the grain may change to one row per rule_id and rule_version. Multiple-rule analysis would require an alert fact or case-to-rule bridge.

2.8 dim_date

Status: Approved Phase 8.1 blueprint.

Business purpose: Provides a consistent calendar hierarchy for daily, weekly, monthly, quarterly and yearly fraud analysis, including MoM, YoY and YTD reporting. Fiscal and public-holiday fields are optional pending business definitions.

Grain: One row per real calendar day, plus controlled Unknown Date and Not Applicable members.

Source: Generated directly in the Gold layer using a dbt SQL model or date-generation macro.

Natural key: full_date for normal calendar rows.

Surrogate key: date_key: YYYYMMDD integer for normal dates; 0 = Unknown Date; -1 = Not Applicable.

Column	Definition
date_key	Integer date surrogate key. YYYYMMDD for real dates; reserved special keys for non-calendar members.
full_date	Actual DATE for real calendar rows; NULL for special members.
date_type	Valid Date, Unknown or Not Applicable.
date_label	Reporting label for the date or special member.
year	Calendar year.
quarter_num	Calendar quarter number 1-4.
quarter_name	Calendar quarter label.
month_year	Month-year reporting label.
month_num	Month number 1-12.
month_name	Month name.
week_of_year	Agreed ISO/calendar week number.
day_of_month	Day number 1-31.
day_name	Day-of-week name.
is_weekend	Boolean weekend indicator; NULL for special members.

Special member	Approved values
Unknown Date	date_key = 0; full_date = NULL; date_type/date_label = Unknown; numeric and Boolean calendar attributes = NULL where not meaningful.
Not Applicable	date_key = -1; full_date = NULL; date_type/date_label = Not Applicable; numeric and Boolean calendar attributes = NULL where not meaningful.

Test area	Required checks
Keys	date_key unique/not null; full_date unique/not null for normal date rows only.
Ranges	month_num 1-12; day_of_month 1-31; quarter_num 1-4; week_of_year within agreed range.
Continuity	For normal rows only: datediff(day, min(full_date), max(full_date)) + 1 = count(*).
Special members	Exactly one date_key = 0 and one date_key = -1 with approved labels.
Fact relationships	All fact date keys reference an existing dim_date.date_key.

Optional future attributes: fiscal_year, fiscal_quarter, fiscal_month_num, is_public_holiday and public_holiday_name after the relevant Australian financial-year and holiday scope are confirmed.

3. Raw-layer tables

Raw-layer status
The following are working physical definitions based on Version 0.1. Exact sheet names, column names and data types must be

reconciled with fraud_scam_raw_sample_data_larger.xlsx before loading to Snowflake.

3.1 raw_scam_cases

Purpose: Stores source-level fraud/scam investigation case records with minimal transformation.

Grain: One raw row per received case record; duplicates may intentionally exist.

Column	Definition
case_id	Source case identifier.
customer_id	Source customer reference.
account_id	Source account reference.
product_id or product_code	Source product reference; final physical naming pending.
channel_code	Source channel reference.
scam_type_code	Source scam type reference.
rule_id	Source primary/originating rule reference.
case_created_date	Raw case creation date.
case_closed_date	Raw case closure date.
loss_occurred_date	Raw financial-loss date.
case_status	Raw case status.
case_outcome	Raw case outcome.
case_priority	Raw case priority.
alert_count	Raw associated alert count.
transaction_count	Raw associated transaction count.
gross_loss_amount	Raw gross loss.
prevented_amount	Raw prevented amount.
recovered_amount	Raw recovered amount.
reimbursed_amount	Raw reimbursed amount.
source_updated_timestamp	Optional source audit timestamp.

Intentional quality issues:

- Duplicate case_id values
- Null or orphaned dimension references
- Invalid rule_id or scam_type_code
- Inconsistent casing and blank strings
- Invalid or reversed dates
- Negative financial values
- Recovered/reimbursed values exceeding gross loss
- Operationally plausible nulls

3.2 raw_customers

Purpose: Stores raw source-system customer records.

Grain: One raw row per received customer record.

Column	Definition
customer_id	Source customer identifier.
customer_type	Raw customer type.
customer_segment	Raw customer segment.
date_of_birth or age	Source value used to derive age_group, if available.
risk_rating	Raw customer risk classification.
state	Raw state or region.
customer_since_date	Raw relationship start date.
customer_status	Raw customer status.
source_updated_timestamp	Optional source audit timestamp.

Intentional quality issues:

- Duplicate customer records
- Mixed casing
- Blank values
- Invalid state abbreviations
- Invalid dates
- Inconsistent risk ratings
- Null descriptive values

3.3 raw_accounts

Purpose: Stores raw individual account records.

Grain: One raw row per received account record.

Column	Definition
account_id	Source account identifier.
customer_id	Source account owner reference.
product_id or product_code	Source product reference.
account_type	Raw account type.
account_status	Raw account status.
account_open_date	Raw account opening date.
account_closed_date	Raw account closure date.
is_joint_account	Raw joint-account indicator.
source_updated_timestamp	Optional source audit timestamp.

Intentional quality issues:

- Duplicate account IDs
- Orphaned customer/product references
- Mixed status values
- Inconsistent Boolean values
- Closure dates before opening dates
- Missing fields

3.4 raw_products

Purpose: Stores raw banking-product reference data.

Grain: One raw row per received product record.

Column	Definition
product_id or product_code	Stable source product identifier; final physical naming pending.
product_name	Raw product name.
product_category	Raw product grouping.
product_status	Raw product lifecycle status.
source_updated_timestamp	Optional source audit timestamp.

Intentional quality issues:

- Duplicate product identifiers
- Missing product names
- Mixed category spelling
- Inconsistent product statuses
- Blank strings

3.5 raw_channels

Purpose: Stores raw banking-channel reference data.

Grain: One raw row per received channel record.

Column	Definition
channel_code	Source channel identifier.
channel_name	Raw channel name.
channel_type	Raw channel classification.
channel_group	Raw higher-level channel grouping.
source_updated_timestamp	Optional source audit timestamp.

Intentional quality issues:

- Duplicate channel codes
- Leading/trailing spaces
- Mixed casing
- Inconsistent classifications
- Missing descriptive values

3.6 raw_scam_types

Purpose: Stores raw scam-type reference data.

Grain: One raw row per received scam type record.

Column	Definition
scam_type_code	Source scam type identifier.
scam_type_name	Raw scam type name.
scam_category	Raw scam category.
scam_description	Raw description.
active_flag	Raw active indicator.
source_updated_timestamp	Optional source audit timestamp.

Intentional quality issues:

- Duplicate scam type codes
- Missing categories/descriptions
- Misspellings
- Mixed casing
- Inconsistent active indicators
- Blank values

3.7 raw_rules

Purpose: Stores raw fraud/scam detection-rule reference data.

Grain: One raw row per received rule record or rule version, depending on the source extract.

Column	Definition
rule_id	Source rule identifier.
rule_name	Raw rule name.
rule_category	Raw rule category.
rule_description	Raw rule description.
is_active	Raw active indicator.
rule_version	Source rule version.
rule_start_date	Rule effective start date.
rule_end_date	Rule end date where applicable.
last_updated_date	Source update date.

Intentional quality issues:

- Duplicate rule records
- Multiple versions
- Mixed category values
- Inconsistent active indicators
- Missing descriptions
- End dates before start dates
- Null or blank fields

4. Silver-layer tables

Silver-layer rule
Silver models clean, standardise, cast, deduplicate and prepare natural keys. They do not create Gold surrogate keys.

4.1 stg_customers

Purpose: Cleans and standardises customer records for dim_customer.

Grain: One row per accepted current customer_id.

Expected transformations:

- Trim identifiers and text
- Standardise casing, state, risk and status values
- Cast dates
- Derive age_group if required
- Convert flags to consistent data types
- Deduplicate customer records
- Retain the most appropriate current record

Output columns:

- customer_id
- customer_type
- customer_segment
- age_group
- risk_rating
- state
- customer_since_date
- customer_status
- Relevant quality flags/audit metadata

Feeds: dim_customer

4.2 stg_accounts

Purpose: Cleans and standardises account records for dim_account and case validation.

Grain: One row per accepted current account_id.

Expected transformations:

- Trim identifiers
- Standardise account types/statuses
- Standardise Boolean values
- Cast account dates
- Flag invalid date sequences
- Deduplicate account records
- Preserve customer/product natural-key references

Output columns:

- account_id
- customer_id
- product_id or product_code
- account_type
- account_status
- account_open_date
- account_closed_date
- is_joint_account
- last_updated_at or audit metadata
- Relevant quality flags

Feeds: dim_account

4.3 stg_products

Purpose: Cleans and standardises product reference data for dim_product.

Grain: One row per accepted product natural key.

Expected transformations:

- Trim and standardise product identifiers
- Standardise names, categories and statuses
- Convert blanks to NULL or approved controlled values
- Deduplicate product records

Output columns:

- Confirmed product natural key: product_id or product_code
- product_name
- product_category
- product_status
- Relevant quality flags/audit metadata

Feeds: dim_product

4.4 stg_channels

Purpose: Cleans and standardises channel reference data for dim_channel.

Grain: One row per accepted standardised channel_code.

Expected transformations:

- Trim channel_code
- Convert codes to consistent casing
- Standardise names, types and groups
- Convert blanks to NULL or approved values
- Deduplicate channel records

Output columns:

- channel_code
- channel_name
- channel_type
- channel_group
- Relevant quality flags/audit metadata

Feeds: dim_channel

4.5 stg_scam_types

Purpose: Cleans and standardises scam-type reference data for dim_scam_type.

Grain: One row per accepted standardised scam_type_code.

Expected transformations:

- Trim and standardise codes
- Correct known name/category variants
- Standardise casing
- Standardise active indicators
- Handle missing category/description values
- Deduplicate records
- Retain optional missing-value flags in Silver

Output columns:

- scam_type_code
- scam_type_name
- scam_category
- scam_description
- active_flag
- Optional data-quality flags
- Relevant audit metadata

Feeds: dim_scam_type

4.6 stg_rules

Purpose: Cleans and standardises rule reference data and selects the current rule definition.

Grain: One row per accepted current rule_id.

Expected transformations:

- Trim and standardise rule IDs
- Standardise categories and active indicators
- Cast and validate dates
- Resolve duplicate records
- Select current/latest rule version
- Retain rule_version as a descriptive attribute

Output columns:

- rule_id
- rule_name
- rule_category
- rule_description
- is_active
- rule_version
- rule_start_date
- rule_end_date
- last_updated_date
- Relevant quality flags/audit metadata

Feeds: dim_rule

4.7 stg_scam_cases

Purpose: Cleans, standardises and deduplicates case records before the Gold fact is built.

Grain: One row per accepted fraud/scam investigation case_id.

Expected transformations:

- Trim and standardise natural-key references
- Reject/quarantine missing case_id
- Deduplicate case_id
- Standardise status, outcome and priority
- Cast and validate dates
- Standardise counts and financial amounts
- Flag or handle invalid values
- Derive case indicators
- Prepare case-level physical measures
- Preserve invalid/unmatched dimension values for Gold mapping and quality review

Output columns:

- case_id
- customer_id
- account_id
- product_id or product_code
- channel_code
- scam_type_code
- rule_id
- case_created_date
- case_closed_date
- loss_occurred_date
- case_status
- case_outcome
- case_priority
- is_confirmed_scam
- is_false_positive
- is_closed_case
- alert_count
- transaction_count
- gross_loss_amount
- prevented_amount
- recovered_amount
- reimbursed_amount
- net_loss_amount
- days_to_resolve
- Relevant quality flags/source metadata

Feeds: fct_scam_cases

5. Source-to-target summary

Raw table	Silver model	Gold target
raw_customers	stg_customers	dim_customer
raw_accounts	stg_accounts	dim_account
raw_products	stg_products	dim_product
raw_channels	stg_channels	dim_channel
raw_scam_types	stg_scam_types	dim_scam_type

Raw table	Silver model	Gold target
raw_rules	stg_rules	dim_rule
raw_scam_cases	stg_scam_cases	fct_scam_cases
Generated calendar	Not applicable	dim_date

6. Decision register and remaining implementation items

Decision	Status	Implementation note
Main fact grain	Approved	One row per fraud/scam investigation case_id.
Gold business surrogate keys	Approved	Deterministic hashes generated from standardised natural keys.
Date keys	Approved	YYYYMMDD for normal dates; 0 Unknown; -1 Not Applicable.
Current-state customer/account dimensions	Approved	SCD Type 2 deferred.
Unknown fact row	Rejected	Missing case_id rows are rejected/quarantined.
No Rule vs Unknown Rule	Approved	Separate NO_RULE and UNK members.
Dimension descriptive date NULLs	Approved	Allowed when genuinely unknown; does not break surrogate-key relationships.
Gold quality flags	Approved principle	Retain primarily in Silver unless business reporting requires them.
Product natural-key physical name	Open	Confirm product_id versus product_code from the workbook and use one name consistently.
Accepted-value domains	Open	Confirm exact case statuses, outcomes, priorities, categories, flags and state values from source/business definitions.
Fiscal calendar	Optional/open	Add only after confirming July-June requirements and labels.
Public holiday calendar	Optional/open	Confirm national versus state-specific scope.
Raw/Silver physical columns	Open	Reconcile with fraud_scam_raw_sample_data_larger.xlsx before implementation.
Unknown-rate thresholds	Phase 9	Define monitoring thresholds after physical build.

6.1 Next implementation sequence

1. Create the GitHub repository and begin version-controlled documentation.
2. Set up Snowflake, dbt, VS Code and required schemas.
3. Load and profile the larger Raw workbook.
4. Reconcile the physical Raw/Silver columns and resolve the product natural-key name.
5. Physically build and test all Silver models.
6. Build Gold dimensions in the approved order.
7. Build fct_scam_cases and protect the one-row-per-case grain.
8. Run structural tests, reconciliation and Phase 9 validation.

Document control: This Version 0.2 document is the approved Phase 8.1 modelling baseline. Any physical implementation change that alters grain, key meaning, metric definitions or special-member handling must be recorded in the modelling decision log and reflected in the next dictionary version.